3-3-5

1. A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

x	-1	0	2	3
$g(x)$	4	1	0	-2
$g'(x)$	-2	-6	-3	-4

The table above gives selected values for a differentiable and decreasing function g and its derivative. Let f be the function with $f(3) = 2$ and derivative given by $f'(x) = x\cos(e^x)$.

(a) Find $f''(\sin 2)$. Express your answer as a decimal approximation.

(b) Let h be the function defined by $h(x) = g(f(2x))$. Find $h'(1.5)$. Express your answer as a decimal approximation.

(c) Write an equation for the line tangent to the graph of g^{-1}, the inverse function of g, at $x = -2$.

(d) The point $(3, 1)$ lies on the curve in the xy-plane given by the equation $(f(x))^2 + g(x) = xy - 1$. What is the value of $\frac{dy}{dx}$ at the point $(3, 1)$? Express your answer as a decimal approximation.

Part A

A decimal approximation is required and must be accurate to three places after the decimal point.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

0	1

The student response accurately includes a correct decimal approximation of $f''(\sin 2)$

Solution:

$f''(\sin 2) = -2.173 \ (\text{or } -2.172)$

Part B

The first and second points require evidence of applying the chain rule twice and no errors. At most 1 out of 3 points is earned for partial communication of chain rule with a maximum of one computational error. Substitution of function values is required. A

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decimal approximation is required and must be accurate to three places after the decimal point.Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2	3

The student response accurately includes all three of the criteria below.

- ☐ chain rule (2 points for full credit)
- ☐ answer

Solution:

$$h'(x) = g'(f(2x)) \cdot f'(2x) \cdot 2$$

$$\begin{aligned} h'(1.5) = g'(f(3)) \cdot f'(3) \cdot 2 &= g'(2) \cdot \left(3\cos\left(e^3\right)\right) \cdot 2 \\ &= -3\left(3\cos\left(e^3\right)\right) \cdot 2 \\ &= -18\cos\left(e^3\right) = -5.915 \ (\text{or } -5.914) \end{aligned}$$

Part C

The second point may be earned if response contains an incorrect value for slope based on one computational error.Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2

The student response accurately includes both of the criteria below.

- ☐ slope
- ☐ equation for tangent line

Solution:

Because $g(3) = -2$, $g^{-1}(-2) = 3$.

$$\left(g^{-1}\right)'(-2) = \frac{1}{g'\left(g^{-1}(-2)\right)} = \frac{1}{g'(3)} = -\frac{1}{4}$$

An equation for the tangent line is $y = 3 - \frac{1}{4}(x + 2)$.

Part D

At most 1 out of 2 implicit differentiation points is earned for correct implicit differentiation with a correct application of the chain rule on only one side of the equation OR a correct application of the chain rule on both sides of the equation with a maximum of one computational error. Substitution of function values is required. The third point requires a decimal approximation that is accurate to three places after the decimal point.Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

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0	1	2	3

The student response accurately includes all three of the criteria below.

- implicit differentiation (2 points for full credit)
- answer

Solution:

$$\frac{d}{dx}\left((f(x))^2 + g(x)\right) = \frac{d}{dx}(xy - 1)$$

$$\Rightarrow \ 2f(x) \cdot f'(x) + g'(x) = y + x\frac{dy}{dx}$$

$$\Rightarrow \ 2f(3) \cdot f'(3) + g'(3) = 1 + 3\frac{dy}{dx}$$

$$\Rightarrow \ 2(2) \cdot \left(3\cos\left(e^3\right)\right) + (-4) = 1 + 3\frac{dy}{dx}$$

$$\Rightarrow \frac{dy}{dx} = -0.352$$

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2. A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

x	-1	0	1	2
$g(x)$	6	4	2	-1
$g'(x)$	-1	-7	-2	-3

The table above gives selected values for a differentiable and decreasing function g and its derivative. Let f be the function with $f(1) = 0$ and derivative given by $f'(x) = x \sin(x^2)$.

(a) Find $f''(e^{1.5})$. Express your answer as a decimal approximation.

(b) Let h be the function defined by $h(x) = f(g(3x))$. Find $h'(0)$. Express your answer as a decimal approximation.

(c) Write an equation for the line tangent to the graph of g^{-1}, the inverse function of g, at $x = -1$.

(d) The point $(1, 4)$ lies on the curve in the xy-plane given by the equation $f(x) + (g(x))^2 = xy$. What is the value of $\frac{dy}{dx}$ at the point $(1, 4)$? Express your answer as a decimal approximation.

Part A

A decimal approximation is required and must be accurate to three places after the decimal point.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

0	1 ✓

The student response accurately includes a correct decimal approximation of $f''(e^{1.5})$

Solution:

$f''(e^{1.5}) = 14.144$

Part B

The first and second points require evidence of applying the chain rule twice and no errors. At most 1 out of 3 points is earned for partial communication of chain rule with a maximum of one computational error. Substitution of function values is required. A

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decimal approximation is required and must be accurate to three places after the decimal point.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2	3

The student response accurately includes all three of the criteria below.

- [ ] chain rule (2 points for full credit)
- [ ] answer

Solution:

$h'(x) = f'(g(3x)) \cdot g'(3x) \cdot 3$

$$\begin{aligned} h'(0) &= f'(g(0)) \cdot g'(0) \cdot 3 = f'(4) \cdot (-7) \cdot 3 \\ &= 4 \sin 16 \cdot (-21) = -84 \sin 16 \\ &= 24.184 \text{ (or } 24.183) \end{aligned}$$

Part C

The second point may be earned if response contains an incorrect value for slope based on one computational error.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2

The student response accurately includes both of the criteria below.

- [ ] slope
- [ ] equation for tangent line

Solution:

Because $g(2) = -1$, $g^{-1}(-1) = 2$.

$$\left(g^{-1}\right)'(-1) = \frac{1}{g'\left(g^{-1}(-1)\right)} = \frac{1}{g'(2)} = \frac{1}{-3}$$

An equation for the tangent line is $y = 2 - \frac{1}{3}(x+1)$.

Part D

At most 1 out of 2 implicit differentiation points is earned for correct implicit differentiation with a correct application of the chain rule on only one side of the equation OR a correct application of the chain rule on both sides of the equation with a maximum of one computational error.

Substitution of function values is required. The third point requires a decimal approximation that is accurate to three places after the decimal point.

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Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2	3

The student response accurately includes all three of the criteria below.

- ☐ implicit differentiation (2 points for full credit)
- ☐ answer

Solution:

$$\frac{d}{dx}\left(f(x) + (g(x))^2\right) = \frac{d}{dx}(xy)$$

$$\Rightarrow\ f'(x) + 2g(x)\cdot g'(x) = y + x\frac{dy}{dx}$$

$$\Rightarrow\ f'(1) + 2g(1)\cdot g'(1) = 4 + \frac{dy}{dx}$$

$$\Rightarrow\ \sin 1 + 2\cdot 2\cdot\ (-2) = 4 + \frac{dy}{dx}$$

$$\Rightarrow \frac{dy}{dx} = \sin 1 - 12 = -11.159\ (\text{or}\ \ -11.158)$$

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3.

x	$f(x)$	$f'(x)$	$g(x)$	$g'(x)$
1	6	4	2	5
2	9	2	3	1
3	10	-4	4	2
4	-1	3	6	7

The functions f and g are differentiable for all real numbers, and g is strictly increasing. The table above gives values of the functions and their first derivatives at selected values of x. The function h is given by $h(x) = f(g(x)) - 6$.
If g^{-1} is the inverse function of g, write an equation for the line tangent to the graph of $y = g^{-1}(x)$ at $x = 2$.

Part D

1 point is earned: g-1(2)

1 point is earned: (g-1)' (2)

1 point is earned for the tangent line equation

$g(1) = 2, so\, g^{-1}(2) = 1$
$\left(g^{-1}\right)'(2) = \frac{1}{g'(g^{-1}(2))} = \frac{1}{g'(1)} = \frac{1}{5}$

An equation of the tangent line is $y - 1 = \frac{1}{5}(x - 2)$.

0	1	2	3

The student response earns all of the following points:

1 point is earned: g-1(2)

1 point is earned: (g-1)' (2)

1 point is earned for the tangent line equation

$g(1) = 2, so\, g^{-1}(2) = 1$
$\left(g^{-1}\right)'(2) = \frac{1}{g'(g^{-1}(2))} = \frac{1}{g'(1)} = \frac{1}{5}$

An equation of the tangent line is $y - 1 = \frac{1}{5}(x - 2)$.

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4. If $y = \sin^{-1}(x-1)$, then $\frac{dy}{dx} =$

(A) $\frac{1}{\sqrt{2-x}}$

(B) $\frac{1}{\sqrt{1-x^2}}$

(C) $\frac{1}{\sqrt{2x-x^2}}$

(D) $\frac{1}{x^2-2x+2}$

Answer C

Correct. Since $\frac{d}{dx}\sin^{-1}x = \frac{1}{\sqrt{1-x^2}}$, the derivative of this composite function using the chain rule is $\frac{d}{dx}\sin^{-1}(x-1) = \frac{1}{\sqrt{1-(x-1)^2}} \cdot \frac{d}{dx}(x-1) = \frac{1}{\sqrt{1-(x^2-2x+1)}} \cdot 1 = \frac{1}{\sqrt{2x-x^2}}$.

5. If $y = \arcsin\left(e^{4x}\right)$, then $\frac{dy}{dx} =$

(A) $\frac{4e^{4x}}{1+e^{8x}}$

(B) $\frac{4e^{4x}}{\sqrt{1-e^{8x}}}$

(C) $\frac{4e^{4x}}{\sqrt{e^{8x}-1}}$

(D) $4e^{4x}\arccos\left(e^{4x}\right)$

Answer B

Correct. The derivative is found using the derivative of the arcsine function and the chain rule. Let $f(x) = \arcsin x$ and $g(x) = e^{4x}$. Then $\arcsin\left(e^{4x}\right) = f(g(x))$. Since $f'(x) = \frac{d}{dx}\arcsin x = \frac{1}{\sqrt{1-x^2}}$, the chain rule gives $\frac{dy}{dx} = f'(g(x))g'(x) = f'\left(e^{4x}\right) \cdot \frac{d}{dx}\left(e^{4x}\right) = \frac{1}{\sqrt{1-\left(e^{4x}\right)^2}} \cdot 4e^{4x} = \frac{4e^{4x}}{\sqrt{1-e^{8x}}}$.

6. Let f be a differentiable function such that $f(3) = 15$, $f(6) = 3$, $f'(3) = -8$, and $f'(6) = -2$. The function g is differentiable and $g(x) = f^{-1}(x)$ for all x. What is the value of $g'(3)$?

(A) $-\frac{1}{2}$

(B) $-\frac{1}{8}$

(C) $\frac{1}{6}$

(D) $\frac{1}{3}$

(E) The value of $g'(3)$ cannot be determined from the information given.

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7. If $f(x) = \arccos\left(x^2\right)$, then f'(x)=

(A) $\frac{1}{\sqrt{1-x^4}}$

(B) $\frac{-2x}{\sqrt{1-x^4}}$ ✓

(C) $\frac{2x}{\sqrt{1-x^4}}$

(D) $\frac{-4x^3}{\sqrt{1-x^4}}$

(E) $\frac{4x^3}{\sqrt{1-x^4}}$

8. If $y = \sin^{-1}(5x)$, then $\frac{dy}{dx} =$

(A) $\frac{1}{1+25x^2}$

(B) $\frac{5}{1+25x^2}$

(C) $\frac{-5}{\sqrt{1-25x^2}}$

(D) $\frac{1}{\sqrt{1-25x^2}}$

(E) $\frac{5}{\sqrt{1-25x^2}}$ ✓

9. If $\arcsin x = \ln y$, then $\frac{dy}{dx} =$

(A) $\frac{y}{\sqrt{1-x^2}}$ ✓

(B) $\frac{xy}{\sqrt{1-x^2}}$

(C) $\frac{y}{1+x^2}$

(D) $e^{\arcsin x}$

(E) $\frac{e^{\arcsin x}}{1+x^2}$

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10.

x	$f(x)$	$g(x)$	$f'(x)$
-4	0	-9	5
-2	4	-7	4
0	6	-4	2
2	7	-3	1
4	10	-2	3

The table above gives values of the differentiable functions *f* and *g*, and f', the derivative of *f*, at selected values of *x*. If $g(x) = f^{-1}(x)$, what is the value of $g'(4)$?

(A) $-\frac{1}{3}$

(B) $-\frac{1}{4}$

(C) $-\frac{3}{100}$

(D) $\frac{1}{4}$ ✓

(E) $\frac{1}{3}$

11. If $\lim\limits_{h\to 0}\frac{\arcsin(a+h)-\arcsin(a)}{h} = 2$, which of the following could be the value of *a?*

(A) $\frac{\sqrt{2}}{2}$

(B) $\frac{\sqrt{3}}{2}$ ✓

(C) $\sqrt{3}$

(D) $\frac{1}{2}$

(E) 2

12. Let *f* be the function defined by $f(x) = x^3 + x$. If $g(x) = f^{-1}(x)$ and $g(2) = 1$, what is the value of $g'(2)$?

(A) $\frac{1}{13}$

(B) $\frac{1}{4}$ ✓

(C) $\frac{7}{4}$

(D) 4

(E) 13

13. Let *f* and *g* be functions that are differentiable everywhere. If *g* is the inverse function of *f* and if $g(-2) = 5$ and f'(5)=−1/2 , then $g'(-2) =$

(A) 2

(B) 1/2

(C) 1/5

(D) $-\frac{1}{5}$

(E) -2 ✓

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14. Let f be the function defined by $f(x) = x^3 + x^2 + x$. Let $g(x) = f^{-1}(x)$, where g(3) = 1. What is the value of g'(3)?

(A) 1/39

(B) 1/34

(C) 1/6 ✓

(D) 1/3

(E) 39

15. The functions f and g are differentiable. For all x, $f(g(x)) = x$ and $g(f(x)) = x$. If $f(3) = 8$ and $f'(3) = 9$, what are the values of $g(8)$ and $g'(8)$?

(A) $g(8) = \frac{1}{3}$ and $g'(8) = -\frac{1}{9}$

(B) $g(8) = \frac{1}{3}$ and $g'(8) = \frac{1}{9}$

(C) $g(8) = 3$ and $g'(8) = -9$

(D) $g(8) = 3$ and $g'(8) = -\frac{1}{9}$

(E) $g(8) = 3$ and $g'(8) = \frac{1}{9}$ ✓

16.

x	$f(x)$	$f'(x)$
0	1	1
1	3	4
2	11	13

The table above gives selected values for a differentiable and increasing function f and its derivative. If g is the inverse function of f, what is the value of $g'(3)$?

(A) $\frac{1}{13}$

(B) $\frac{1}{4}$ ✓

(C) 1

(D) 4

(E) 13

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17.

x	$f(x)$	$f'(x)$
0	49	0
1	2	–8
2	–1	–80

The table above gives selected values for a differentiable and decreasing function f and its derivative. If f^{-1} is the inverse function of f, what is the value of $(f^{-1})'(2)$?

(A) -80

(B) $-\frac{1}{8}$ ✓

(C) $-\frac{1}{80}$

(D) $\frac{1}{80}$

(E) $\frac{1}{8}$

18. If $f(x) = \sin^{-1}x$, then $f'\left(\frac{\sqrt{3}}{2}\right) =$

(A) $\frac{\pi}{6}$

(B) $\frac{\pi}{3}$

(C) $\frac{4}{7}$

(D) 2 ✓

Answer D

This option is correct. The derivative of the inverse sine function $\sin^{-1} x$ is $\frac{1}{\sqrt{1-x^2}}$. Since $f(x) = \sin^{-1}x$,

$$f'\left(\frac{\sqrt{3}}{2}\right) = \frac{1}{\sqrt{1-\left(\frac{\sqrt{3}}{2}\right)^2}} = \frac{1}{\sqrt{\frac{1}{4}}} = 2.$$

19. $\frac{d}{dx}\left(\tan^{-1}x + 2\sqrt{x}\right) =$

(A) $-\frac{1}{\sin^2 x} + \frac{1}{\sqrt{x}}$

(B) $\frac{1}{\sqrt{1-x^2}} - 4\sqrt[3]{x}$

(C) $\frac{1}{\sqrt{1-x^2}} + \frac{1}{\sqrt{x}}$

(D) $\frac{1}{1+x^2} - 4\sqrt[3]{x}$

(E) $\frac{1}{1+x^2} + \frac{1}{\sqrt{x}}$ ✓

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20. If $y = \arctan(\cos x)$, then $\frac{dy}{dx} =$

(A) $\frac{-\sin x}{1+\cos^2 x}$

(B) $-(\operatorname{arcsec}(\cos x))^2 \sin x$

(C) $(\operatorname{arcsec}(\cos x))^2$

(D) $\frac{1}{(\arccos x)^2+1}$

(E) $\frac{1}{1+\cos^2 x}$

21. If $f(x) = \sin x + 2x + 1$ and g is the inverse function of f, what is the value of $g'(1)$?

(A) $\frac{1}{3}$

(B) 1

(C) 3

(D) $\frac{1}{2+\cos 1}$

(E) $2 + \cos 1$

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22. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Let f be an increasing function with $f(0) = 2$. The derivative of f is given by $f'(x) = \sin(\pi x) + x^2 + 3$.

(a) Find $f''(-2)$.

(b) Write an equation for the line tangent to the graph of $y = \frac{1}{f(x)}$ at $x = 0$.

(c) Let g be the function defined by $g(x) = f\left(\sqrt{3x^2+4}\right)$. Find $g'(2)$.

(d) Let h be the inverse function of f. Find $h'(2)$.

Part A

The second point cannot be earned without the first point. The second point does not require a simplified answer; trigonometric function values do not need to be evaluated.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2 ✓

The student response accurately includes both of the criteria below.

- ☐ chain rule
- ☐ answer

Solution:

$f''(x) = \pi\cos(\pi x) + 2x$

$f''(-2) = \pi\cos(-2\pi) + 2(-2) = \pi - 4$

Part B

The second point may be earned based on a derivative expression that includes application of power rule and chain rule with a maximum of one computational error.

The third point may be earned if response contains a slope that is consistent with the derivative expression.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

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0	1	2	3

The student response accurately includes all three of the criteria below.

- ☐ $\frac{d}{dx}\left(\frac{1}{f(x)}\right)$
- ☐ slope
- ☐ equation for tangent line

Solution:

At $x = 0$, $y = \frac{1}{f(0)} = \frac{1}{2}$.

$$\frac{d}{dx}\left(\frac{1}{f(x)}\right) = \frac{d}{dx}(f\ (x))^{-1} = -(f\ (x))^{-2} f'\ (x) = -\frac{f'(x)}{(f(x))^2}$$

$$\frac{d}{dx}\left(\frac{1}{f(x)}\right)\Big|_{x=0} = -\frac{f'(0)}{(f(0))^2} = -\frac{3}{4}$$

An equation for the tangent line is $y = \frac{1}{2} - \frac{3}{4}x$.

Part C

The first and second points require evidence of applying the chain rule twice and no errors. At most 1 out of 3 points is earned for partial communication of chain rule with a maximum of one computational error. Substitution of function values is required. A simplified answer is not required.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

0	1	2	3

The student response accurately includes all three of the criteria below.

- ☐ chain rule (2 points for full credit)
- ☐ answer

Solution:

$$g'\ (x) = f'\left(\sqrt{3x^2+4}\right) \cdot \frac{6x}{2\sqrt{3x^2+4}} = f'\left(\sqrt{3x^2+4}\right) \cdot \frac{3x}{\sqrt{3x^2+4}}$$

$$g'\ (2) = f'\left(\sqrt{3\ \cdot\ 2^2+4}\right) \cdot \frac{3 \cdot 2}{\sqrt{3 \cdot 2^2+4}} = f'\ (4) \cdot \frac{3}{2} = 19 \cdot \frac{3}{2} = \frac{57}{2}$$

Part D

Substitution of function values is required.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

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0	1

The student response accurately includes a correct value of $h'(2)$

Solution:

Because $f(0) = 2$, $h(2) = 0$.

$h'(2) = \frac{1}{f'(h(2))} = \frac{1}{f'(0)} = \frac{1}{3}$

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23. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Let f be an increasing function with $f(0)=3$. The derivative of f is given by $f'(x)=\cos(\pi x)+x^4+6$.

(a) Find $f''(-3)$.

(b) Write an equation for the line tangent to the graph of $y=(f(x))^2$ at $x=0$.

(c) Let g be the function defined by $g(x)=f\left(\sqrt{2x^2+7}\right)$. Find $g'(3)$.

(d) Let h be the inverse function of f. Find $h'(3)$.

Part A

The second point cannot be earned without the first point. The second point does not require a simplified answer; trigonometric function values do not need to be evaluated.Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2 ✓

The student response accurately includes both of the criteria below.

- [ ] chain rule
- [ ] answer

Solution:

$$f''(x)=-\pi\sin(\pi x)+4x^3$$

$$f''(-3)=-\pi\sin(-3\pi)+4(-3)^3=-108$$

Part B

The second point may be earned based on a derivative expression that includes application of power rule and chain rule with a maximum of one computational error. Trigonometric function values do not need to be evaluated.The third point may be earned if response contains a slope that is consistent with the derivative expression.Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2	3 ✓

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The student response does not accurately include any of the criteria below.

- ☐ $\frac{d}{dx}\left((f(x))^2\right)$
- ☐ slope
- ☐ equation for tangent line

Solution:

At $x = 0$, $y = (f(0))^2 = 3^2 = 9$.

$$\frac{d}{dx}\left((f(x))^2\right) = 2f(x)f'(x)$$

$$\left.\frac{d}{dx}\left((f(x))^2\right)\right|_{x=0} = 2f(0)f'(0) \quad = 2(3)\ (\cos 0 + 6) = 6(\cos 0 + 6) = 42$$

An equation for the tangent line is y equals, $y = 9 + 42x$.

Part C

The first and second points require evidence of applying the chain rule twice and no errors. At most 1 out of 3 points is earned for partial communication of chain rule with a maximum of one computational error. Substitution of function values is required. Trigonometric function values do not need to be evaluated. A simplified answer is not required.Select a point value to view scoring criteria, solutions, and/or examples to score the response.

0	1	2	3 ✓

The student response accurately includes all three of the criteria below.

- ☐ chain rule (2 points for full credit)
- ☐ answer

Solution:

$$g'(x) = f'\left(\sqrt{2x^2+7}\right) \cdot \frac{4x}{2\sqrt{2x^2+7}} = f'\left(\sqrt{2x^2+7}\right) \cdot \frac{2x}{\sqrt{2x^2+7}}$$

$$g'(3) = f'\left(\sqrt{2 \cdot 3^2 + 7}\right) \cdot \frac{2 \cdot 3}{\sqrt{2 \cdot 3^2 + 7}} = f'(5) \cdot \frac{6}{5} = (\cos(5\pi) + 631) \cdot \frac{6}{5} = (630)\frac{6}{5} = 756$$

Part D

Substitution of function values is required. Trigonometric function values do not need to be evaluated.Select a point value to view scoring criteria, solutions, and/or examples to score the response.

0	1 ✓

The student response accurately includes the criteria below.

3-3-5

- [ ] answer

Solution:

Because $f(0) = 3$, $h(3) = 0$.

$h'(3) = \frac{1}{f'(h(3))} = \frac{1}{f'(0)} = \frac{1}{(\cos 0+6)} = \frac{1}{7}$

24. Let $f(x) = (2x+1)^3$ and let g be the inverse function of f. Given that $f(0) = 1$, what is the value of $g'(1)$?

(A) $-\frac{2}{27}$

(B) $\frac{1}{54}$

(C) $\frac{1}{27}$

(D) $\frac{1}{6}$ ✓

(E) 6

25. What is the slope of the line tangent to the curve $y = \arctan(4x)$ at the point at which $x = \frac{1}{4}$?

(A) 2 ✓

(B) $\frac{1}{2}$

(C) 0

(D) $-\frac{1}{2}$

(E) -2

26. An equation for a tangent to the graph of $y = \arcsin\frac{x}{2}$ at the origin is

(A) $x - 2y = 0$ ✓

(B) $x - y = 0$

(C) $x = 0$

(D) $y = 0$

(E) $\pi x - 2y = 0$

27. Let f be the function defined by $f(x) = 2x + e^x$. If $g(x) = f^{-1}(x)$ for all x and the point (0,1) is on the graph of f, what is the value of $g'(1)$?

(A) $\frac{1}{2+e}$

(B) $\frac{1}{3}$ ✓

(C) $\frac{1}{2}$

(D) 3

(E) $2+e$

3-3-5

28. The function h is given by $h(x) = x^5 + 3x - 2$ and $h(1) = 2$. If h^{-1} is the inverse of h, what is the value of $\left(h^{-1}\right)'(2)$?

(A) $\frac{1}{83}$

(B) $\frac{1}{8}$ ✓

(C) $\frac{1}{2}$

(D) 1

(E) 8

29. The function $y = g(x)$ is differentiable and increasing for all real numbers. On what intervals is the function $y = g\left(x^3 - 6x^2\right)$ increasing?

(A) $(-\infty, 0]$ and $[4, \infty)$ only ✓

(B) $[0, 4]$ only

(C) $[2, \infty)$ only

(D) $[6, \infty)$ only

(E) $(-\infty, \infty)$

3-3-5

30. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Consider the curve given by the equation $2xy + y^2 = 8$ for $y > 0$.

(a) Show that $\frac{dy}{dx} = \frac{-y}{x+y}$.

(b) Write an equation for the line tangent to the curve at the point $(1, 2)$.

(c) Evaluate $\frac{d^2y}{dx^2}$ at the point $(1, 2)$.

(d) The points $(1, 2)$ and $\left(\frac{7}{2}, 1\right)$ are on the curve. Find the value of $\left(y^{-1}\right)'(1)$.

Part A

The first point is earned for $2x\frac{dy}{dx} + 2y + 2y\frac{dy}{dx} = 0$ or equivalent

The second point cannot be earned without the first point. The second point is earned for a response that arrives at the given expression rather than an algebraically equivalent expression.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2 ✓

The student response accurately includes both of the criteria below.

- ☐ implicit differentiation
- ☐ verification

Solution:

$$\frac{d}{dx}\left(2xy + y^2\right) = \frac{d}{dx}(8)$$
$$\Rightarrow \quad 2x\frac{dy}{dx} + 2y + 2y\frac{dy}{dx} = 0$$
$$\Rightarrow \quad x\frac{dy}{dx} + y\frac{dy}{dx} = -y$$
$$\Rightarrow \quad \frac{dy}{dx} = \frac{-y}{x+y}$$

Part B

The second point may be earned if response contains an incorrect value for slope based on one computational error.

3-3-5

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2

The student response accurately includes both of the criteria below

- ☐ slope
- ☐ equation for tangent line

Solution:

$$\left.\frac{dy}{dx}\right|_{(x,\ y)=(1,\ 2)} = \frac{-2}{1+2} = -\frac{2}{3}$$

An equation for the tangent line is $y = 2 - \frac{2}{3}(x-1)$.

Part C

The first point is earned with

$$\frac{(x+y)\cdot\left(-1\frac{dy}{dx}\right)-(-y)\left(1+1\frac{dy}{dx}\right)}{(x+y)^2}$$ or equivalent.

The second point is earned with an algebraic or numerical substitution for $\frac{dy}{dx}$.

The response does not require an explicit expression for $\frac{d^2y}{dx^2}$.

The third point does not require a simplified answer.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

0	1	2	3

The student responses accurately includes all three of the criteria below.

- ☐ implicit differentiation (2 points for full credit)
- ☐ answer

Solution:

$$\frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{-y}{x+y}\right)$$

$$= \frac{(x+y)\cdot\left(-1\frac{dy}{dx}\right)-(-y)\left(1+1\frac{dy}{dx}\right)}{(x+y)^2}$$

$$= \frac{(x+y)\cdot\left(\frac{y}{x+y}\right)+y\left(1+\left(\frac{-y}{x+y}\right)\right)}{(x+y)^2} = \frac{y^2+2xy}{(x+y)^3}$$

3-3-5

$$\left.\frac{d^2y}{dx^2}\right|_{(x,\ y)=(1,\ 2)}=\frac{2^2+2(1)(2)}{(1+2)^3}=\frac{8}{27}$$

Part D

The second point may be earned if response contains an incorrect value based on one computational error. Substitution of function values is required.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2

The student response does not accurately include any of the criteria below.

- [ ] $\frac{dy}{dx}$ at $\left(\frac{7}{2},\ 1\right)$
- [ ] answer

Solution:

$$\left.\frac{dy}{dx}\right|_{(x,\ y)=\left(\frac{7}{2},\ 1\right)}=\frac{-1}{\left(\frac{7}{2}+1\right)}=-\frac{2}{9}$$

Because $\left(\frac{7}{2},\ 1\right)$ is on the curve, $\left(1,\ \frac{7}{2}\right)$ is on the inverse.

$$\left(y^{-1}\right)'(1)=\frac{1}{y'\left(y^{-1}(1)\right)}=\frac{1}{y'\left(\frac{7}{2}\right)}=-\frac{9}{2}$$

$$\left(y^{-1}\right)'(1)=\frac{1}{y'\left(y^{-1}(1)\right)}=\frac{1}{y'\left(\frac{7}{2}\right)}=-\frac{9}{2}$$

3-3-5

31. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Consider the curve given by the equation $(2y+1)^3 - 24x = -3$.

(a) Show that $\frac{dy}{dx} = \frac{4}{(2y+1)^2}$.

(b) Write an equation for the line tangent to the curve at the point $(-1, -2)$.

(c) Evaluate $\frac{d^2y}{dx^2}$ at the point $(-1, -2)$.

(d) The point $\left(\frac{1}{6}, 0\right)$ is on the curve. Find the value of $\left(y^{-1}\right)'(0)$.

Part A

The first point is earned for $3(2y+1)^2 \cdot 2\frac{dy}{dx} - 24 = 0$ or equivalent.

The second point cannot be earned without the first point. The second point is earned for a response that arrives at the given expression rather than an algebraically equivalent expression.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2 ✓

The student response accurately includes both of the criteria below.

- ☐ implicit differentiation
- ☐ verification

Solution:

$$\frac{d}{dx}\left((2y+1)^3 - 24x\right) = \frac{d}{dx}(-3)$$

$$\Rightarrow\ 3(2y+1)^2 \cdot 2\frac{dy}{dx} - 24 = 0$$

$$\Rightarrow\ \frac{dy}{dx} = \frac{24}{6(2y+1)^2} = \frac{4}{(2y+1)^2}$$

Part B

The second point may be earned if response contains an incorrect value for slope based on one computational error.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

3-3-5

0	1	2

The student response accurately includes both of the criteria below.

- ☐ slope
- ☐ equation for tangent line

Solution:

$$\frac{dy}{dx}\bigg|_{(x,\ y)=(-1,\ -2)} = \frac{4}{(2(-2)+1)^2} = \frac{4}{9}$$

An equation for the tangent line is $y = -2 + \frac{4}{9}(x+1)$.

Part C

The first point is earned with $4\,(-2)(2y+1)^{-3} \cdot 2\frac{dy}{dx}$ or equivalent.

The second point is earned with an algebraic or numerical substitution for $\frac{dy}{dx}$. The response does not require an explicit expression for $\frac{d^2y}{dx^2}$. The third point does not require a simplified answer.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

0	1	2	3

The student responses accurately includes all three of the criteria below.

- ☐ implicit differentiation (2 points for full credit)
- ☐ answer

Solution:

$$\frac{d^2y}{dx^2} = \frac{d}{dx}\left(4(2y+1)^{-2}\right)$$

$$= 4\,(-2)(2y+1)^{-3} \cdot 2\frac{dy}{dx}$$

$$= \frac{-16}{(2y+1)^3} \cdot \frac{4}{(2y+1)^2} = \frac{-64}{(2y+1)^5}$$

$$\frac{d^2y}{dx^2}\bigg|_{(x,\ y)=(-1,\ -2)} = \frac{-64}{(2(-2)+1)^5} = \frac{-64}{-243} = \frac{64}{243}$$

Part D

The second point may be earned if response contains an incorrect value based on one computational error. Substitution of function values is required.

3-3-5

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2

The student response accurately includes both of the criteria below.

- [ ] $\frac{dy}{dx}$ at $\left(\frac{1}{6},\ 0\right)$
- [ ] answer

Solution:

$$\left.\frac{dy}{dx}\right|_{(x,\ y)=\left(\frac{1}{6},\ 0\right)} = \frac{4}{(2\cdot 0+1)^2} = 4$$

Because $\left(\frac{1}{6},\ 0\right)$ is on the curve, $\left(0,\ \frac{1}{6}\right)$ is on the inverse.

$$\left(y^{-1}\right)'(0) = \frac{1}{y'\left(y^{-1}(0)\right)} = \frac{1}{y'\left(\frac{1}{6}\right)} = \frac{1}{4}$$

32. Let $f$and g be inverse functions that are differentiable for all x. If $f(3) = -2$ and $g'(-2) = -4$, which of the following statements must be false?

I. $f'(0) = \frac{1}{4}$
II. $f'(3) = -\frac{1}{4}$
III. $f'(5) = -\frac{1}{4}$

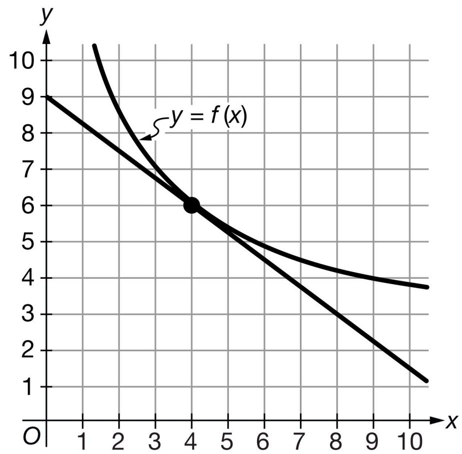
(A) I only
(B) II only
(C) III only
(D) I and III only

Answer A

Correct. If the derivative of g changed signs at some value of x, then g would have a relative minimum or a relative maximum at that value of x. In either case, g could not have an inverse function. Therefore, the derivative of g cannot change signs. Since $g'(-2)$ is negative, $g'(x)$ must be negative for all values of x. Then $f'(x) = \frac{1}{g'(f(x))} < 0$ for all x. Therefore, statement I must be false.
Statement II is true. Because f and g are inverse functions and $f(3) = -2$, it follows that $f'(3) = \frac{1}{g'(f(3))} = \frac{1}{g'(-2)} = -\frac{1}{4}$.
Statement III could be true. For example, if $f(x) = -\frac{1}{4}(x-3) - 2$ and $g(x) = -4(x+2) + 3$, then f and g are inverses, $f(3) = -2$, and $g'(-2) = 4$. In addition, $f'(5) = -\frac{1}{4}$.

3-3-5

33.



The graph of the decreasing differentiable function f is shown above. Also shown is the line tangent to the graph of f at the point $(4, 6)$. Let g be the inverse of f. Which of the following statements about g' is true?

(A) $g'(4) = -\frac{4}{3}$

(B) $g'(4) = -\frac{3}{4}$

(C) $g'(6) = -\frac{4}{3}$

(D) $g'(6) = -\frac{3}{4}$

Answer C

Correct. This statement can be confirmed using the chain rule and the definition of an inverse function. The line tangent to the graph of f at the point $(4, 6)$ has slope $f'(4) = -\frac{3}{4}$. Since $g(x) = f^{-1}(x)$, then $g(6) = 4$ and $f(g(x)) = x$. By the chain rule, $\frac{d}{dx}f(g(x)) = f'(g(x))g'(x) = \frac{d}{dx}(x) = 1 \Rightarrow g'(6) = \frac{1}{f'(g(6))} = \frac{1}{f'(4)} = \frac{1}{-\frac{3}{4}} = -\frac{4}{3}$.

34. Let f be the increasing function defined by $f(x) = x^3 + 2x^2 + 4x + 5$, where $f(-1) = 2$. If g is the inverse function of f, which of the following is a correct expression for $g'(2)$?

(A) $g'(2) = \frac{1}{f'(2)}$

(B) $g'(2) = \frac{1}{f'(-1)}$

(C) $g'(2) = f'(-1)$

(D) $g'(2) = f'(2)$

Answer B

Correct. This can be confirmed using the chain rule and the definition of an inverse function. Since $f(g(x)) = x$, it follows that $\frac{d}{dx}f(g(x)) = f'(g(x))g'(x) = \frac{d}{dx}(x) = 1 \Rightarrow g'(x) = \frac{1}{f'(g(x))}$. Therefore, $g'(2) = \frac{1}{f'(g(2))} = \frac{1}{f'(-1)}$.

3-3-5

35.

x	0	2	4
$f(x)$	8	5	2
$f'(x)$	-1	-2	-5

The table above gives selected values for a differentiable and decreasing function f and its derivative. If $g(x) = f^{-1}(x)$ for all x, which of the following is a correct expression for $g'(2)$?

(A) $g'(2) = f'(2) = -2$

(B) $g'(2) = \frac{1}{f'(2)} = -\frac{1}{2}$

(C) $g'(2) = \frac{1}{f'(4)} = -\frac{1}{5}$ ✓

(D) $g'(2) = -\frac{f'(2)}{(f(2))^2} = \frac{2}{25}$

Answer C

Correct. This value can be confirmed using the chain rule and the definition of an inverse function. Since $f(g(x)) = x$, it follows that $\frac{d}{dx} f(g(x)) = f'(g(x))g'(x) = \frac{d}{dx}(x) = 1 \Rightarrow g'(x) = \frac{1}{f'(g(x))}$.
Therefore, $g'(2) = \frac{1}{f'(g(2))} = \frac{1}{f'(4)} = -\frac{1}{5}$.

36. An increasing function f satisfies $f(10) = 5$ and $f'(10) = 8$. Which of the following statements about the inverse of f must be true?

(A) $\left(f^{-1}\right)'(5) = 10$

(B) $\left(f^{-1}\right)'(8) = 10$

(C) $\left(f^{-1}\right)'(5) = 8$

(D) $\left(f^{-1}\right)'(5) = \frac{1}{8}$ ✓

Answer D

Correct. This statement can be confirmed using the chain rule and the definition of an inverse function. $f^{-1}(5) = 10$ and $f(f^{-1}(x)) = x$. By the chain rule, $\frac{d}{dx} f(f^{-1}(x)) = f'(f^{-1}(x))\left(f^{-1}\right)'(x) = \frac{d}{dx}(x) = 1 \Rightarrow \left(f^{-1}\right)'(5) = \frac{1}{f'(f^{-1}(5))} = \frac{1}{f'(10)} = \frac{1}{8}$.

3-3-5

37. For which of the following increasing functions f does $\left(f^{-1}\right)'(20) = \frac{1}{5}$?

(A) $f(x) = x + 5$

(B) $f(x) = x^3 + 5x + 20$ ✓

(C) $f(x) = x^5 + 5x + 14$

(D) $f(x) = e^x + 5x + 19$

Answer B

Correct. $f(x) = 20$ when $x = 0$. Therefore $f^{-1}(20) = 0$. By the chain rule, $\left(f^{-1}\right)'(20) = \frac{1}{f'(f^{-1}(20))} = \frac{1}{f'(0)} = \frac{1}{3x^2+5}\Big|_{x=0} = \frac{1}{5}$.

38. Let f and g be inverse functions that are differentiable for all x. If $f(-5) = 7$ and $g'(7) = 3$, which of the following statements must be false?

I. $f'(3) = -\frac{1}{3}$
II. $f'(-5) = \frac{1}{3}$
III. $f'(7) = \frac{1}{3}$

(A) I only ✓

(B) II only

(C) III only

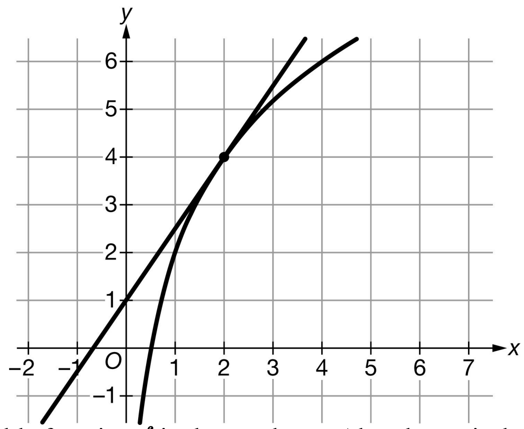
(D) I and III only

Answer A

Correct. If the derivative of g changed signs at some value of x, then g would have a relative minimum or a relative maximum at that value of x. In either case, g could not have an inverse function. Therefore, the derivative of g cannot change signs. Since $g'(7)$ is positive, $g'(x)$ must be positive for all values of x. Then $f'(x) = \frac{1}{g'(f(x))} > 0$ for all x. Therefore, statement I must be false.
Statement II is true. Because f and g are inverse functions and $f(-5) = 7$, it follows that $f'(-5) = \frac{1}{g'(f(-5))} = \frac{1}{g'(7)} = \frac{1}{3}$.
Statement III could be true. For example, if $f(x) = \frac{1}{3}(x+5) + 7$ and $g(x) = 3(x-7) - 5$, then f and g are inverses, $f(-5) = 7$, and $g'(7) = 3$. In addition, $f'(7) = \frac{1}{3}$.

3-3-5

39.



The graph of the increasing differentiable function f is shown above. Also shown is the line tangent to the graph of f at the point $(2, 4)$. Let g be the inverse of f. Which of the following statements about g' is true?

(A) $g'(2) = \frac{2}{3}$

(B) $g'(2) = \frac{3}{2}$

(C) $g'(4) = \frac{2}{3}$ ✓

(D) $g'(4) = \frac{3}{2}$

Answer C

Correct. This statement can be confirmed using the chain rule and the definition of an inverse function. The line tangent to the graph of f at the point $(2, 4)$ has slope $f'(2) = \frac{3}{2}$. Since $g(x) = f^{-1}(x)$, then $g(4) = 2$ and $f(g(x)) = x$. By the chain rule, $\frac{d}{dx} f(g(x)) = f'(g(x))g'(x) = \frac{d}{dx}(x) = 1 \Rightarrow g'(4) = \frac{1}{f'(g(4))} = \frac{1}{f'(2)} = \frac{1}{\frac{3}{2}} = \frac{2}{3}$.

40. Let f be the decreasing function defined by $f(x) = -x^3 - 6x^2 - 12x + 8$, where $f(4) = -8$. If g is the inverse function of f, which of the following is a correct expression for $g'(-8)$?

(A) $g'(-8) = \frac{1}{f'(-8)}$

(B) $g'(-8) = \frac{1}{f'(4)}$ ✓

(C) $g'(-8) = f'(4)$

(D) $g'(-8) = f'(-8)$

Answer B

Correct. This can be confirmed using the chain rule and the definition of an inverse function. Since $f(g(x)) = x$, it follows that $\frac{d}{dx} f(g(x)) = f'(g(x))g'(x) = \frac{d}{dx}(x) = 1 \Rightarrow g'(x) = \frac{1}{f'(g(x))}$. Therefore, $g'(-8) = \frac{1}{f'(g(-8))} = \frac{1}{f'(4)}$.

3-3-5

41.

x	-4	0	3
$f(x)$	0	3	5
$f'(x)$	1	2	4

The table above gives selected values for a differentiable and increasing function f and its derivative. If $g(x) = f^{-1}(x)$ for all x, which of the following is a correct expression for $g'(0)$?

(A) $g'(0) = f'(0) = 2$

(B) $g'(0) = \frac{1}{f'(0)} = \frac{1}{2}$

(C) $g'(0) = \frac{1}{f'(-4)} = 1$ ✓

(D) $g'(0) = -\frac{f'(0)}{(f(0))^2} = -\frac{2}{9}$

Answer C

Correct. This value can be confirmed using the chain rule and the definition of an inverse function. Since $f(g(x)) = x$, it follows that $\frac{d}{dx} f(g(x)) = f'(g(x))g'(x) = \frac{d}{dx}(x) = 1 \Rightarrow g'(x) = \frac{1}{f'(g(x))}$.
Therefore, $g'(0) = \frac{1}{f'(g(0))} = \frac{1}{f'(-4)} = \frac{1}{1} = 1$.

42. A decreasing function g satisfies $g(4) = 6$ and $g'(4) = -2$. Which of the following statements about the inverse of g must be true?

(A) $\left(g^{-1}\right)'(6) = 4$

(B) $\left(g^{-1}\right)'(-2) = 4$

(C) $\left(g^{-1}\right)'(6) = -2$

(D) $\left(g^{-1}\right)'(6) = -\frac{1}{2}$ ✓

Answer D

Correct. This statement can be confirmed using the chain rule and the definition of an inverse function. $g^{-1}(6) = 4$ and $g(g^{-1}(x)) = x$.
By the chain rule,
$\frac{d}{dx} g(g^{-1}(x)) = g'(g^{-1}(x))\left(g^{-1}\right)'(x) = \frac{d}{dx} x = 1 \Rightarrow \left(g^{-1}\right)'(x) = \frac{1}{g'(g^{-1}(x))} \Rightarrow \left(g^{-1}\right)'(6) = \frac{1}{g'(g^{-1}(6))} = \frac{1}{g'(4)} = -\frac{1}{2}$.

3-3-5

43. For which of the following decreasing functions f does $\left(f^{-1}\right)'(10) = -\frac{1}{8}$?

(A) $f(x) = -5x + 15$

(B) $f(x) = -2x^3 - 2x + 14$ ✓

(C) $f(x) = -x^5 - 4x + 15$

(D) $f(x) = e^{-2x} - x + 9$

Answer B

Correct. $f(x) = 10$ when $x = 1$. Therefore, $f^{-1}(10) = 1$. By the chain rule, $\left(f^{-1}\right)'(10) = \frac{1}{f'(f^{-1}(10))} = \frac{1}{f'(1)} = \frac{1}{-6x^2-2}\Big|_{x=1} = -\frac{1}{8}$.

44. Which of the following expressions can be differentiated using the product rule?

(A) $\cos\left(\sqrt{x}\right)$

(B) $x^2 \tan^{-1} x$ ✓

(C) $x^4 + \arcsin x$

(D) $\left(8x^3 - 5x + 2\right)^{\pi}$

Answer B

Correct. This expression is the product of x^2 and $\tan^{-1} x$, and therefore requires use of the product rule to differentiate the expression.

45. Which of the following requires the use of implicit differentiation to find $\frac{dy}{dx}$?

(A) $y - x^2 - 3x + 5 = 0$

(B) $y = \ln(3 + x) + x^2$

(C) $y = \ln(y + x) + x^2$ ✓

(D) $y = \frac{x^3-4}{3x+2}$

Answer C

Correct. In this expression y is an implicitly-defined function and the independent variable x cannot be isolated. Therefore implicit differentiation is needed to find the derivative $\frac{dy}{dx}$.

3-3-5

46. Which of the following could be used to find the slope of the line tangent to the curve $\sin^{-1}(2x^2+y^2)=\frac{2}{x}+y^2$?

(A) $\frac{1}{\sqrt{1-(2x^2+y^2)^2}}=\frac{-2}{x^2}+2y\frac{dy}{dx}$

(B) $\frac{4x+2y}{\sqrt{1-(2x^2+y^2)^2}}=\frac{-2}{x^2}+2y$

(C) $\frac{4x+2y\frac{dy}{dx}}{\sqrt{1-(2x^2+y^2)^2}}=\frac{-2}{x^2}+2y\frac{dy}{dx}$

(D) $\frac{4x+2y\frac{dy}{dx}}{\sqrt{1-(2x^2+y^2)}}=\frac{-2}{x^2}+2y\frac{dy}{dx}$

Answer C

Correct. The chain rule must be used, not only for the implicit differentiation but also because the inverse sine function has an inside component.

$$\frac{4x+2y\frac{dy}{dx}}{\sqrt{1-(2x^2+y^2)^2}}=\frac{-2}{x^2}+2y\frac{dy}{dx}$$

47. Which of the following does not require the use of the chain rule to find $\frac{dy}{dx}$?

(A) $y=\cos^{-1}(10x^5-x^2)$

(B) $4x^{10}+6y^3=x^2y^5-2$

(C) $y=10\sqrt{x}-\frac{4}{x^3}+x^5$

(D) $\sin(2x-y)+e^{2y}+\frac{x}{6}=0$

Answer C

Correct. $\frac{dy}{dx}$ can be found using the power rule. There is no composite function involved, so no chain rule is necessary.

3-3-5

48.

x	4	8
$f(x)$	11	6
$f'(x)$	-4	-3

The table above gives selected values for a differentiable and decreasing function f and its derivative. Let g be the decreasing function given by $g(x) = f(4x) - f(2x)$, where $g(2) = f(8) - f(4) = -5$. Which of the following describes a correct process for finding $\left(g^{-1}\right)'(-5)$?

(A) $\left(g^{-1}\right)'(-5) = \frac{1}{g'(g^{-1}(-5))} = \frac{1}{g'(2)}$ and $g'(2) = 4f'(8) - 2f'(4)$ ✓

(B) $\left(g^{-1}\right)'(-5) = \frac{1}{g'(g^{-1}(-5))} = \frac{1}{g'(2)}$ and $g'(2) = f'(8) - f'(4)$

(C) $\left(g^{-1}\right)'(-5) = g'(g^{-1}(-5)) = g'(2)$ and $g'(2) = f'(8) - f'(4)$

(D) $\left(g^{-1}\right)'(-5) = g'(g^{-1}(-5)) = g'(2)$ and $g'(2) = 4f'(8) - 2f'(4)$

Answer A

Correct. The appropriate rule for the derivative of the inverse of g is $\left(g^{-1}\right)'(-5) = \frac{1}{g'(g^{-1}(-5))} = \frac{1}{g'(2)}$. By the chain rule, $g'(x) = 4f'(4x) - 2f'(2x)$. Therefore, $g'(2) = 4f'(8) - 2f'(4)$.

49. For which of the following functions would the quotient rule be considered the best method for finding the derivative?

(A) $y = (2x+1)^{-\frac{1}{2}}$

(B) $y = \frac{2x+1}{x}$

(C) $y = \sin^{-1}(2x+1)$

(D) $y = \frac{\sin(2x+1)}{2x+1}$ ✓

Answer D

Correct. This function cannot be rewritten in a way that would simplify the differentiation. The quotient rule is the best method.

3-3-5

50. Which of the following expressions can be differentiated using the product rule?

(A) $\arcsin(\cos x)$

(B) $\sin x\,(\arccos x)$

(C) $e^x + \arctan x$

(D) $\left(12x^2 + 3x - 6\right)^e$

Answer B

Correct. This expression is the product of $\sin x$ and $\arccos x$. Therefore, differentiation of the expression requires the use of the product rule.

51. For which of the following functions would the quotient rule be considered the best method for finding the derivative?

(A) $y = \left(x^3 + x\right)^{-2}$

(B) $y = \frac{x^3+x}{x}$

(C) $y = \cos^{-1}(x^3 + x)$

(D) $y = \frac{\cos(x^3+x)}{x^3+x}$

Answer D

Correct. This function cannot be rewritten in a way that would simplify the differentiation. The quotient rule is the best method.

52. Which of the following requires the use of implicit differentiation to find $\frac{dy}{dx}$?

(A) $2y + 3x^2 - x = 5$

(B) $y = e^{3+x} + x^3$

(C) $y = e^{y+x} + x^3$

(D) $y = \frac{x^4+3}{4x^3-2}$

Answer C

Correct. In this expression y is an implicitly defined function and the independent variable x cannot be isolated. Therefore, implicit differentiation is needed to find the derivative $\frac{dy}{dx}$.

3-3-5

53. Which of the following could be used to find the slope of the line tangent to the curve $\tan^{-1}(x-2y+2)=x^2-3y+\tan^{-1}(2)-1$?

(A) $\frac{1}{1+(x-2y+2)^2}=2x-3\frac{dy}{dx}$

(B) $\frac{-1}{1+(x-2y+2)^2}=2x-3$

(C) $\frac{1-2\frac{dy}{dx}}{1+(x-2y+2)^2}=2x-3\frac{dy}{dx}$ ✓

(D) $\frac{1-2\frac{dy}{dx}}{1+(x-2y+2)}=2x-3\frac{dy}{dx}$

Answer C

Correct. The chain rule must be used, not only for the implicit differentiation but also because the inverse tangent function has an “inside” component.

$\frac{1-2\frac{dy}{dx}}{1+(x-2y+2)^2}=2x-3\frac{dy}{dx}$

54. Which of the following does not require the use of the chain rule to find $\frac{dy}{dx}$?

(A) $y=\sin^{-1}(3x^2-4)$

(B) $3x^6-4y^2=2xy^5+7$

(C) $y=3x^2-\sqrt{x}+\frac{2}{x}$ ✓

(D) $\cos(x+y)+2^y-x=0$

Answer C

Correct. $\frac{dy}{dx}$ can be found using the power rule. There is no composite function involved, so no chain rule is necessary.

3-3-5

55.

x	3	6
$f(x)$	4	5
$f'(x)$	$\frac{1}{3}$	$\frac{3}{4}$

The table above gives selected values for a differentiable and increasing function f and its derivative. Let g be the increasing function given by $g(x) = f(x) + f(2x)$, where $g(3) = f(3) + f(6) = 9$. Which of the following describes a correct process for finding $(g^{-1})'(9)$?

(A) $(g^{-1})'(9) = \frac{1}{g'(g^{-1}(9))} = \frac{1}{g'(3)}$ and $g'(3) = f'(3) + 2f'(6)$ ✓

(B) $(g^{-1})'(9) = \frac{1}{g'(g^{-1}(9))} = \frac{1}{g'(3)}$ and $g'(3) = f'(3) + f'(6)$

(C) $(g^{-1})'(9) = g'(g^{-1}(9)) = g'(3)$ and $g'(3) = f'(3) + f'(6)$

(D) $(g^{-1})'(9) = g'(g^{-1}(9)) = g'(3)$ and $g'(3) = f'(3) + 2f'(6)$

Answer A

Correct. The appropriate rule for the derivative of the inverse of g is $(g^{-1})'(9) = \frac{1}{g'(g^{-1}(9))} = \frac{1}{g'(3)}$. By the chain rule, $g'(x) = f'(x) + 2f'(2x)$. Therefore,
$g'(3) = f'(3) + 2f'(6)$.

56.

x	$f(x)$	$f'(x)$
1	2	3
2	3	4
3	5	2

The function f is increasing and differentiable. Selected values of f and its derivative f' are given in the table above. What is the value of $\left(f^{-1}\right)'(3)$?

(A) $\frac{1}{4}$ ✓

(B) $\frac{1}{2}$

(C) 1

(D) 2

Answer A

Correct. Since $f\left(f^{-1}(x)\right) = x$, the chain rule can be used to determine that $f'\left(f^{-1}(x)\right)\left(f^{-1}\right)'(x) = 1$. Substituting $x = 3$ gives the following.
$1 = f'\left(f^{-1}(3)\right)\left(f^{-1}\right)'(3) \Rightarrow \left(f^{-1}\right)'(3) = \frac{1}{f'(f^{-1}(3))}$
Because $f(2) = 3$, then $f^{-1}(3) = 2$. Therefore, $\left(f^{-1}\right)'(3) = \frac{1}{f'(2)} = \frac{1}{4}$.

3-3-5

57. $\frac{d}{dx}\left(\sin^{-1}x\right)\Big|_{x=\frac{1}{2}} =$

(A) $\frac{1}{1+\left(\frac{1}{2}\right)^2}$

(B) $\frac{1}{\sqrt{1-\left(\frac{1}{2}\right)^2}}$ ✓

(C) $\cos^{-1}\left(\frac{1}{2}\right)$

(D) $-\csc\left(\frac{1}{2}\right)\cot\left(\frac{1}{2}\right)$

Answer B

Correct. If $y=\sin^{-1}x$, then $\sin y = x$. Using the chain rule to differentiate this implicit relation gives $\cos y\frac{dy}{dx} = 1 \Rightarrow \frac{dy}{dx} = \frac{1}{\cos y}$. Since $y=\sin^{-1}x$ implies that $-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$, the value of $\cos y$ is nonnegative. Then $\cos^2 y = 1-\sin^2 y = 1-x^2 \Rightarrow \cos y = \sqrt{1-x^2}$. Therefore, $\frac{dy}{dx} = \frac{1}{\cos y} = \frac{1}{\sqrt{1-x^2}}$. This is evaluated at $x=\frac{1}{2}$.

58. $\frac{d}{dx}\left(\csc^{-1}\left(e^x\right)\right) =$

(A) $-e^x\cdot\cot(e^x)\cdot\csc\left(e^x\right)$

(B) $e^x\cdot\cot\left(e^x\right)\cdot\csc^3\left(e^x\right)$

(C) $\frac{e^x}{\sqrt{1-e^{2x}}}$

(D) $\frac{-1}{e^x\sqrt{1-\left(\frac{1}{e^{2x}}\right)}}$ ✓

Answer D

Correct. If $y=\csc^{-1}x$, then $\csc y = x$ or $\frac{1}{\sin y} = x$. Using the chain rule to differentiate this implicit relation gives $\frac{-\cos y}{\sin^2 y}\cdot\frac{dy}{dx} = 1 \Rightarrow \frac{dy}{dx} = -\frac{\sin^2 y}{\cos y}$. Since $\frac{1}{\sin y} = x$, then $\sin y = \frac{1}{x}$ and $\cos y = \sqrt{1-\sin^2 y}$, it follows that $\frac{dy}{dx} = \frac{-\sin^2 y}{\cos y} = \frac{-\left(\frac{1}{x}\right)^2}{\sqrt{1-\left(\frac{1}{x}\right)^2}} = \frac{-1}{x^2\sqrt{1-\left(\frac{1}{x}\right)^2}}$.

Therefore, by the chain rule, $\frac{d}{dx}\left(\csc^{-1}\left(e^x\right)\right) = \frac{-1}{e^{2x}\cdot\sqrt{1-\left(\frac{1}{e^x}\right)^2}}\cdot e^x = \frac{-1}{e^x\cdot\sqrt{1-\left(\frac{1}{e^{2x}}\right)}}$.

3-3-5

59. $\frac{d}{dx}\left(\sec^{-1}x\right) =$

(A) $-\sin x$

(B) $\tan x \ \sec x$

(C) $\frac{1}{x^2\sqrt{1-\left(\frac{1}{x}\right)^2}}$

(D) $\frac{-1}{x^2\sqrt{1-\left(\frac{1}{x}\right)^2}}$

Answer C

Correct. If $y = \sec^{-1}x$, then $\sec y = x$ or $\frac{1}{\cos y} = x$. Using the chain rule to differentiate this implicit relation gives $\frac{\sin y}{\cos^2 y} \cdot \frac{dy}{dx} = 1 \Rightarrow \frac{dy}{dx} = \frac{\cos^2 y}{\sin y}$. Since $\cos y = \frac{1}{x}$ and $\sin y = \sqrt{1-\cos^2 x} = \sqrt{1-\left(\frac{1}{x}\right)^2}$, $\frac{dy}{dx} = \frac{\left(\frac{1}{x}\right)^2}{\sqrt{1-\left(\frac{1}{x}\right)^2}} = \frac{1}{x^2\sqrt{1-\left(\frac{1}{x}\right)^2}}$.

60. If $f(x) = \arcsin x$, then $\lim\limits_{x\to\frac{1}{2}} \frac{f(x)-f\left(\frac{1}{2}\right)}{x-\frac{1}{2}}$ is

(A) 0

(B) $\frac{\pi}{6}$

(C) $\frac{2}{\sqrt{3}}$

(D) nonexistent

Answer C

Correct. The expression in the limit is the definition of the derivative of $\arcsin x$ at $x = \frac{1}{2}$. The value of the limit is therefore $\frac{d}{dx}\arcsin x\Big|_{x=\frac{1}{2}} = \frac{1}{\sqrt{1-x^2}}\Big|_{x=\frac{1}{2}} = \frac{1}{\sqrt{\frac{3}{4}}} = \frac{2}{\sqrt{3}}$.

3-3-5

61. $\frac{d}{dx}\left(\tan^{-1}(3x)\right) =$

(A) $3\sec^2(3x)$

(B) $-3\csc^2(3x)$

(C) $\frac{3}{\sqrt{1-(3x)^2}}$

(D) $\frac{3}{1+(3x)^2}$

Answer D

Correct. If $y = \tan^{-1} x$, then $\tan y = x$. Using the chain rule to differentiate this implicit relation gives $\sec^2 y \frac{dy}{dx} = 1 \Rightarrow \frac{dy}{dx} = \frac{1}{\sec^2 y}$.
Since
$\sec^2 y = \frac{1}{\cos^2 y} = \frac{\cos^2 y + \sin^2 y}{\cos^2 y} = 1 + \frac{\sin^2 y}{\cos^2 y} = 1 + \tan^2 y,$
it follows that $\frac{dy}{dx} = \frac{1}{\sec^2 y} = \frac{1}{1+\tan^2 y} = \frac{1}{1+x^2}$.
Therefore, by the chain rule, $\frac{d}{dx}\left(\tan^{-1}(3x)\right) = \frac{1}{1+(3x)^2} \cdot 3 = \frac{3}{1+(3x)^2}$.

62. $\frac{d}{dx}\left(\cot^{-1} x\right) =$

(A) $-\csc^2 x$

(B) $\sec^2 x$

(C) $-\frac{1}{1+x^2}$

(D) $\frac{1}{1+x^2}$

Answer C

Correct. If $y = \cot^{-1} x$, then $\cot y = x$. Using the chain rule to differentiate this implicit relation gives
$-\csc^2 y \frac{dy}{dx} = 1 \Rightarrow \frac{dy}{dx} = -\frac{1}{\csc^2 y}$. Since $\csc^2 y = \frac{1}{\sin^2 y} = \frac{\sin^2 y + \cos^2 y}{\sin^2 y} = 1 + \frac{\cos^2 y}{\sin^2 y} = 1 + \cot^2 y,$
$\frac{dy}{dx} = -\frac{1}{\csc^2 y} = -\frac{1}{1+\cot^2 y} = -\frac{1}{1+x^2}$.

3-3-5

63. If $f(x) = \arctan x$, then $\lim_{x \to \sqrt{3}} \frac{f(x) - f\left(\sqrt{3}\right)}{x - \sqrt{3}}$ is

(A) 0

(B) $\frac{1}{4}$ ✓

(C) $\frac{\pi}{3}$

(D) nonexistent

Answer B

Correct. The expression in the limit is the definition of the derivative of $\arctan x$ at $x = \sqrt{3}$. The value of the limit is therefore $\frac{d}{dx} \arctan x \Big|_{x=\sqrt{3}} = \frac{1}{1+x^2}\Big|_{x=\sqrt{3}} = \frac{1}{1+3} = \frac{1}{4}$.

64. $\frac{d}{dx}\left(\sin^{-1}\left(x^2\right)\right)\Big|_{x=\frac{1}{4}} =$

(A) $\frac{2\left(\frac{1}{4}\right)}{1+\left(\frac{1}{4}\right)^4}$

(B) $\frac{2\left(\frac{1}{4}\right)}{\sqrt{1-\left(\frac{1}{4}\right)^4}}$ ✓

(C) $2\left(\frac{1}{4}\right)\cos^{-1}\left(\frac{1}{16}\right)$

(D) $-2\left(\frac{1}{4}\right)\cot\left(\frac{1}{16}\right)\csc\left(\frac{1}{16}\right)$

Answer B

Correct. If $y = \sin^{-1}\left(x^2\right)$, then $\sin y = x^2$. Using the chain rule to differentiate this implicit relation gives $\cos y \frac{dy}{dx} = 2x \Rightarrow \frac{dy}{dx} = \frac{2x}{\cos\ y}$. Since $y = \sin^{-1}\left(x^2\right)$ implies that $0 \leq y \leq \frac{\pi}{2}$, the value of $\cos\ y$ is nonnegative. Then $\cos^2 y = 1 - \sin^2 y = 1 - x^4 \Rightarrow \cos\ y = \sqrt{1 - x^4}$. Therefore, $\frac{dy}{dx} = \frac{2x}{\cos\ y} = \frac{2x}{\sqrt{1-x^4}}$. This is evaluated at $x = \frac{1}{4}$.

3-3-5

65. $\frac{d}{dx}\left(\cos^{-1}(-3x)\right) =$

(A) $\frac{3}{\sqrt{1-(-3x)^2}}$ ✓

(B) $\frac{-3}{\sqrt{1-(-3x)^2}}$

(C) $-\sin^{-1}(-3x)\cdot(-3)$

(D) $-\cos^{-2}(-3x)\cdot(-3)$

Answer A

Correct. If $y = \cos^{-1}(-3x)$, then $\cos\ y = -3x$. Using the chain rule to differentiate this implicit relation gives $-\sin y\frac{dy}{dx} = -3 \Rightarrow \frac{dy}{dx} = \frac{3}{\sin y}$. Since $y = \cos^{-1}(-3x)$ implies that $0 \le y \le \pi$, the value of $\sin y$ is nonnegative. Then $\sin^2 y = 1 - \cos^2 y = 1 - (-3x)^2 \Rightarrow \sin y = \sqrt{1-(-3x)^2}$. Therefore, $\frac{dy}{dx} = \frac{3}{\sin y} = \frac{3}{\sqrt{1-(-3x)^2}}$.

66. Which of the following methods can be used to find the derivative of $y = \arccos\left(\sqrt{x}\right)$ with respect to x ?

I. Use the quotient rule to differentiate $\frac{1}{\cos\left(\sqrt{x}\right)}$.
II. Use the chain rule to differentiate $\cos\left(\arccos\left(\sqrt{x}\right)\right) = \sqrt{x}$.
III. Use implicit differentiation to differentiate the function y in the relation $\cos y = \sqrt{x}$ with respect to x.

(A) I only

(B) III only

(C) II and III only ✓

(D) I, II, and III

Answer C

Correct. The definition of the inverse cosine function means that $\cos\left(\arccos\left(\sqrt{x}\right)\right) = \sqrt{x}$. Using method II, $\frac{d}{dx}\cos\left(\arccos\left(\sqrt{x}\right)\right) = -\sin\left(\arccos\left(\sqrt{x}\right)\right)\cdot\left(\frac{d}{dx}\arccos\left(\sqrt{x}\right)\right) = \frac{d}{dx}\left(\sqrt{x}\right) = \frac{1}{2\sqrt{x}} \Rightarrow \frac{d}{dx}\left(\arccos\left(\sqrt{x}\right)\right) = \frac{-1}{2\sqrt{x}\sin\left(\arccos\left(\sqrt{x}\right)\right)}$. If $\theta = \arccos\left(\sqrt{x}\right)$, then $0 \le \theta \le \frac{\pi}{2}$ and $\cos\ \theta = \sqrt{x}$. Therefore, $\sin\ \theta$ is nonnegative and $\sin^2\theta = 1 - \cos^2\theta \Rightarrow \sin\theta = \sqrt{1-\cos^2\theta} = \sqrt{1-\left(\sqrt{x}\right)^2} = \sqrt{1-x}$ because $0 \le x \le 1$.

The derivative is then $\frac{d}{dx}\arccos\left(\sqrt{x}\right) = \frac{-1}{2\sqrt{x}\cdot\sin\left(\arccos\left(\sqrt{x}\right)\right)} = \frac{-1}{2\sqrt{x}\cdot\sin\ \theta} = \frac{-1}{2\sqrt{x}\sqrt{1-x}} = \frac{-1}{2\sqrt{x(1-x)}}$ for $0 < x < 1$.

Using method III, $-\sin\ y\frac{dy}{dx} = \frac{1}{2\sqrt{x}} \Rightarrow \frac{dy}{dx} = \frac{-1}{2\sqrt{x}\ \sin\ y}$. Since $y = \arccos\left(\sqrt{x}\right)$ implies that $0 \le y \le \frac{\pi}{2}$, the value of $\cos y$ is nonnegative. Then, $\sin^2 y = 1 - \cos^2 y = 1 - x \Rightarrow \sin\ y = \sqrt{1-x}$. Therefore, $\frac{d}{dx}\left(\text{arc}\cos\left(\sqrt{x}\right)\right) = \frac{-1}{2\sqrt{x}\sqrt{1-x}} = \frac{-1}{2\sqrt{x(1-x)}}$ for $0 < x < 1$.

Method I cannot be used since $\frac{1}{\cos\left(\sqrt{x}\right)}$ is not equal to $\arccos\left(\sqrt{x}\right)$, the inverse cosine function.

3-3-5

67. $\frac{d}{dx}\left(\cos^{-1}x\right) =$

(A) $-\frac{1}{\sqrt{1-x^2}}$ ✓

(B) $\frac{1}{\sqrt{1-x^2}}$

(C) $-\sin^{-1}x$

(D) $-\cos^{-2}x$

Answer A

Correct. If $y = \cos^{-1}x$, then $\cos y = x$. Using the chain rule to differentiate this implicit relation gives $-\sin y\frac{dy}{dx} = 1 \Rightarrow \frac{dy}{dx} = -\frac{1}{\sin y}$. Since $y = \cos^{-1}x$ implies that $0 \le y \le \pi$, the value of $\sin y$ is nonnegative. Then $\sin^2 y = 1 - \cos^2 y = 1 - x^2 \Rightarrow \sin y = \sqrt{1-x^2}$. Therefore, $\frac{dy}{dx} = -\frac{1}{\sin y} = -\frac{1}{\sqrt{1-x^2}}$.

68. Which of the following methods can be used to find the derivative of $y = \arcsin x$ with respect to x ?

I. Use the quotient rule to differentiate $\frac{1}{\sin x}$.
II. Use the chain rule to differentiate $\sin(\arcsin x) = x$.
III. Use implicit differentiation to differentiate the function y in the relation $\sin y = x$ with respect to x.

(A) I only

(B) III only

(C) II and III only ✓

(D) I, II, and III

Answer C

Correct. The definition of the inverse sine function means that $\sin(\arcsin x) = x$. Using method II, $\frac{d}{dx}\sin(\arcsin x) = \cos(\arcsin x)\cdot\left(\frac{d}{dx}\arcsin x\right) = \frac{d}{dx}(x) = 1 \Rightarrow \frac{d}{dx}\arcsin x = \frac{1}{\cos(\arcsin x)}$. If $\theta = \arcsin x$, then $-\frac{\pi}{2} \le \theta \le \frac{\pi}{2}$ and $\sin\theta = x$. Therefore, $\cos\theta$ is nonnegative and $\cos^2\theta = 1 - \sin^2\theta = 1 - x^2 \Rightarrow \cos\theta = \sqrt{1-x^2}$. The derivative is then $\frac{d}{dx}\arcsin x = \frac{1}{\cos(\arcsin x)} = \frac{1}{\cos\theta} = \frac{1}{\sqrt{1-x^2}}$ for $-1 < x < 1$.

Using method III, $\cos y\frac{dy}{dx} = 1 \Rightarrow \frac{dy}{dx} = \frac{1}{\cos y}$. Since $y = \arcsin x$ implies that $-\frac{\pi}{2} \le y \le \frac{\pi}{2}$, the value of $\cos y$ is nonnegative. Then, $\cos^2 y = 1 - \sin^2 y = 1 - x^2 \Rightarrow \cos y = \sqrt{1-x^2}$. Therefore, $\frac{d}{dx}(\arcsin x) = \frac{dy}{dx} = \frac{1}{\cos y} = \frac{1}{\sqrt{1-x^2}}$ for $-1 < x < 1$.

Method I cannot be used since $\frac{1}{\sin x}$ is not equal to $\arcsin x$, the inverse sine function.

3-3-5

69. Let f be the function defined by $f(x) = x^3 + x$. If g is the inverse function of f, what is the slope of the line tangent to the graph of g at the point $(2, 1)$?

(A) $-\frac{1}{4}$

(B) $-\frac{1}{13}$

(C) $\frac{1}{13}$

(D) $\frac{1}{4}$

Answer D

Correct. The slope of the line tangent to the graph of g at the point $(2, 1)$ is $g'(2)$. Since g is the inverse of f, $f(g(x)) = x$ for all x. By the chain rule, $f'(g(x))g'(x) = 1$, where $f'(x) = 3x^2 + 1$. Therefore, $g\prime(2) = \frac{1}{f'(g(2))} = \frac{1}{f'(1)} = \frac{1}{4}$.

70.

x	$f(x)$	$g(x)$	$f'(x)$
1	2	0	$\frac{\pi}{2}$
2	π	1	$\frac{\pi}{3}$
π	4	2	$\frac{\pi}{4}$

Let f and g be differentiable functions, where g is the inverse of f. Selected values of f, g, and f' are shown in the table. What is the value of $g'(\pi)$?

(A) $\frac{\pi}{4}$

(B) $\frac{3}{\pi}$

(C) $\frac{\pi}{3}$

(D) $\frac{4}{\pi}$

Answer B

Correct. Since g is the inverse of f, $f(g(x)) = x$ for all x. By the chain rule, $f'(g(x))g'(x) = 1$. Therefore, $g\prime(\pi) = \frac{1}{f'(g(\pi))} = \frac{1}{f'(2)} = \frac{1}{\frac{\pi}{3}} = \frac{3}{\pi}$.

3-3-5

71.

x	$f(x)$	$f'(x)$	$g(x)$	$g'(x)$
1	6	4	2	5
2	9	2	3	1
3	10	−4	4	2
4	−1	3	6	7

The functions f and g are differentiable for all real numbers, and g is strictly increasing. The table above gives values of the functions and their first derivatives at selected values of x. The function h is given by $h(x) = f(g(x)) - 6$.

(a) Explain why there must be a value r for $1 < r < 3$ such that $h(r) = -5$.

(b) Explain why there must be a value c for $1 < c < 3$ such that $h'(c) = -5$.

(c) Let w be the function given by $w(x) = \int_1^{g(x)} f(t)dt$. Find the value of $w'(3)$.

(d) If g^{-1} is the inverse function of g, write an equation for the line tangent to the graph of $y = g^{-1}(x)$ at $x = 2$.

Part A

1 point is earned for $h(1)$ and $h(3)$

1 point is earned for the conclusion, using IVT

$h(1) = f(g(1)) - 6 = f(2) - 6 = 9 - 6 = 3$

$h(3) = f(g(3)) - 6 = f(4) - 6 = -1 - 6 = -7$

Since $h(3) < -5 < h(1)$ and h is continuous, by the Intermediate Value Theorem, there exists a value r, $1 < r < 3$, such that $h(r) = -5$.

0	1	2

The student response earns all of the following points:

1 point is earned for $h(1)$ and $h(3)$

1 point is earned for the conclusion, using IVT

$h(1) = f(g(1)) - 6 = f(2) - 6 = 9 - 6 = 3$

$h(3) = f(g(3)) - 6 = f(4) - 6 = -1 - 6 = -7$

Since $h(3) < -5 < h(1)$ and h is continuous, by the Intermediate Value Theorem, there exists a value r, $1 < r < 3$, such that $h(r) = -5$.

Part B

1 point is earned for: $\dfrac{h(3) - h(1)}{3 - 1}$

1 point is earned for the conclusion, using MVT

3-3-5

$$\frac{h(3)-h(1)}{3-1}=\frac{-7-3}{3-1}=-5$$

Since h is continuous and differentiable, by the Mean Value Theorem, there exists a value c, $1 < c < 3$, such that $h'(c) = -5$.

0	1	2

The student response earns all of the following points:

1 point is earned for: $\dfrac{h(3)-h(1)}{3-1}$

1 point is earned for the conclusion, using MVT

$$\frac{h(3)-h(1)}{3-1}=\frac{-7-3}{3-1}=-5$$

Since h is continuous and differentiable, by the Mean Value Theorem, there exists a value c, $1 < c < 3$, such that $h'(c) = -5$.

Part C

1 point is earned for applying chain rule

1 point is earned for correct answer

$$w'(3)=f(g(3))\cdot g'(3)=f(4)\cdot 2=-2$$

0	1	2

The student response earns all of the following points:

1 point is earned for applying chain rule

1 point is earned for correct answer

$$w'(3)=f(g(3))\cdot g'(3)=f(4)\cdot 2=-2$$

Part D

1 point is earned for $g^{-1}(2)$

1 point is earned for $(g^{-1})'(2)$

3-3-5

1 point is earned for the tangent line equation

$g(1) = 2, \text{so } g^{-1}(2) = 1.$

$\left(g^{-1}\right)'(2) = \frac{1}{g'(g^{-1}(2))} = \frac{1}{g'(1)} = \frac{1}{5}$

An equation of the tangent line is $y - 1 = \frac{1}{5}(x - 2)$.

0	1	2	3

The student response earns all of the following points:

1 point is earned for $g^{-1}(2)$

1 point is earned for $\left(g^{-1}\right)'(2)$

1 point is earned for the tangent line equation

$g(1) = 2, \text{so } g^{-1}(2) = 1.$

$\left(g^{-1}\right)'(2) = \frac{1}{g'(g^{-1}(2))} = \frac{1}{g'(1)} = \frac{1}{5}$

An equation of the tangent line is $y - 1 = \frac{1}{5}(x - 2)$.